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## \\Research\\Investigators\\Frederick\\extravals\\localizer\_64ch\_uncombined

TA: 14 sec Coil Selection: Manual Voxel Size: 0.5×0.5×7.0 mm<sup>3</sup> Acc:: None Rel. SNR: 1.00**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | Off                |
| Wait for User to Start                        | Off                |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | On                 |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Disable auto transfer to PACS                 | Off                |
| Load Images to Stamp Segments                 | On                 |
| Load Images to Graphic Segments               | On                 |
| Graphic segment                               | All Segments       |
| Inline Movie                                  | Off                |

**Routine**

|                     |                    |
|---------------------|--------------------|
| Slice Group         | 1                  |
| Slices              | 1                  |
| Distance Factor     | 20 %               |
| Position            | L0.0 A20.0 H0.0 mm |
| Orientation         | Sagittal           |
| Phase Encoding Dir. | A >> P             |
| Slice Group         | 2                  |
| Slices              | 1                  |
| Distance Factor     | 20 %               |
| Position            | L0.0 A20.0 H0.0 mm |
| Orientation         | Sagittal           |
| Phase Encoding Dir. | A >> P             |
| Slice Group         | 3                  |
| Slices              | 1                  |
| Distance Factor     | 20 %               |
| Position            | L0.0 A20.0 H0.0 mm |
| Orientation         | Sagittal           |
| Phase Encoding Dir. | A >> P             |
| Phase Oversampling  | 0 %                |
| FOV Read            | 250 mm             |
| FOV Phase           | 100.0 %            |
| Slice Thickness     | 7.0 mm             |
| TR                  | 8.6 ms             |
| TE                  | 4.00 ms            |
| Averages            | 2                  |
| Concatenations      | 3                  |
| AutoAlign           | ---                |
| Coil Elements       | HC1-7;NC1,2        |

**Contrast - Common**

|                   |         |
|-------------------|---------|
| TR                | 8.6 ms  |
| TE                | 4.00 ms |
| TD                | 0.00 ms |
| MTC               | Off     |
| Magn. Preparation | None    |

**Contrast - Common**

|                    |           |
|--------------------|-----------|
| Flip Angle         | 20 deg    |
| Fat-Water Contrast | Standard  |
| Dark Blood         | Off       |
| Contrasts          | 1         |
| SWI                | Off       |
| Reconstruction     | Magnitude |

**Contrast - Dynamic**

|                 |                  |
|-----------------|------------------|
| Dynamic Mode    | Standard         |
| Measurements    | 1                |
| Multiple Series | Each Measurement |

**Resolution - Common**

|                  |         |
|------------------|---------|
| FOV Read         | 250 mm  |
| FOV Phase        | 100.0 % |
| Slice Thickness  | 7.0 mm  |
| Base Resolution  | 256     |
| Phase Resolution | 91 %    |
| Interpolation    | On      |

**Resolution - Acceleration**

|                       |         |
|-----------------------|---------|
| Acceleration Mode     | None    |
| Deep Resolve          | Off     |
| Phase Partial Fourier | Off     |
| Asymmetric Echo       | Allowed |

**Resolution - Filter**

|                       |         |
|-----------------------|---------|
| Raw Filter            | Off     |
| Elliptical Filter     | On      |
| Distortion Correction | 2D      |
| Normalize             | Prescan |
| Image Filter          | Off     |

**Geometry - Common**

|                     |                    |
|---------------------|--------------------|
| Slice Group         | 1                  |
| Slices              | 1                  |
| Distance Factor     | 20 %               |
| Position            | L0.0 A20.0 H0.0 mm |
| Orientation         | Sagittal           |
| Phase Encoding Dir. | A >> P             |
| Slice Group         | 2                  |
| Slices              | 1                  |
| Distance Factor     | 20 %               |
| Position            | L0.0 A20.0 H0.0 mm |
| Orientation         | Sagittal           |
| Phase Encoding Dir. | A >> P             |
| Slice Group         | 3                  |
| Slices              | 1                  |
| Distance Factor     | 20 %               |
| Position            | L0.0 A20.0 H0.0 mm |

**Geometry - Common**

|                     |             |
|---------------------|-------------|
| Orientation         | Sagittal    |
| Phase Encoding Dir. | A >> P      |
| Phase Oversampling  | 0 %         |
| FOV Read            | 250 mm      |
| FOV Phase           | 100.0 %     |
| Slice Thickness     | 7.0 mm      |
| TR                  | 8.6 ms      |
| Multi-Slice Mode    | Sequential  |
| Series              | Interleaved |
| Concatenations      | 3           |

**Geometry - AutoAlign**

|                     |                    |
|---------------------|--------------------|
| Slice Group         | 1                  |
| Position            | L0.0 A20.0 H0.0 mm |
| Orientation         | Sagittal           |
| Phase Encoding Dir. | A >> P             |
| Slice Group         | 2                  |
| Position            | L0.0 A20.0 H0.0 mm |
| Orientation         | Sagittal           |
| Phase Encoding Dir. | A >> P             |
| Slice Group         | 3                  |
| Position            | L0.0 A20.0 H0.0 mm |
| Orientation         | Sagittal           |
| Phase Encoding Dir. | A >> P             |
| AutoAlign           | ---                |
| Initial Position    | L0.0 A20.0 H0.0    |
| L                   | 0.0 mm             |
| A                   | 20.0 mm            |
| H                   | 0.0 mm             |
| Initial Orientation | Sagittal           |
| Initial Rotation    | 0.00 deg           |

**Geometry - Saturation**

|                    |          |
|--------------------|----------|
| Saturation Mode    | Standard |
| Special Saturation | None     |

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 0 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**System - Miscellaneous**

|                     |                |
|---------------------|----------------|
| Coil Selection      | Manual         |
| Radial Sorting      | Off            |
| MSMA                | S - C - T      |
| Sagittal            | R >> L         |
| Coronal             | A >> P         |
| Transversal         | F >> H         |
| Coil Combination    | Sum of Squares |
| Matrix Optimization | Off            |

**System - Adjustments**

|                     |          |
|---------------------|----------|
| Adjustment Strategy | Standard |
|---------------------|----------|

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| B0 Shim               | Tune up  |
| B1 Shim               | TrueForm |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | Off      |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |             |
|-------------|-------------|
| Position    | Isocenter   |
| Orientation | Transversal |
| Rotation    | 0.00 deg    |
| A >> P      | 263 mm      |
| R >> L      | 350 mm      |
| F >> H      | 350 mm      |
| Reset       | Off         |

**System - pTx**

|              |            |
|--------------|------------|
| B1 Shim      | TrueForm   |
| Excitation   | Slice-sel. |
| LR Balancing | Off        |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.249470 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Correction Factor   | 1.00           |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                 |        |
|-----------------|--------|
| 1st Signal/Mode | None   |
| TR              | 8.6 ms |
| Segments        | 1      |
| Concatenations  | 3      |

**Physio - Cardiac**

|                    |          |
|--------------------|----------|
| Fat-Water Contrast | Standard |
| Magn. Preparation  | None     |
| Dark Blood         | Off      |
| FOV Read           | 250 mm   |
| FOV Phase          | 100.0 %  |
| Phase Resolution   | 91 %     |

**Physio - PACE**

|                |     |
|----------------|-----|
| Resp. Control  | Off |
| Concatenations | 3   |

**Inline - Liver**

|                      |     |
|----------------------|-----|
| Liver Registration   | Off |
| Save Original Images | On  |

**Inline - Subtraction**

|              |     |
|--------------|-----|
| Subtract     | Off |
| Measurements | 1   |
| StdDev       | Off |

**Inline - Subtraction**

|                      |    |
|----------------------|----|
| Save Original Images | On |
|----------------------|----|

**Inline - MIP**

|                      |     |
|----------------------|-----|
| MIP Sag              | Off |
| MIP Cor              | Off |
| MIP Tra              | Off |
| MIP Time             | Off |
| Radial MIP           | Off |
| Save Original Images | On  |
| MPR Sag              | Off |
| MPR Cor              | Off |
| MPR Tra              | Off |

**Inline - Soft Tissue**

|              |     |
|--------------|-----|
| Wash-in      | Off |
| Wash-out     | Off |
| TTP          | Off |
| PEI          | Off |
| MIP Time     | Off |
| Measurements | 1   |

**Inline - Composing**

|                  |     |
|------------------|-----|
| Inline Composing | Off |
|------------------|-----|

**Sequence - Part 1**

|                   |            |
|-------------------|------------|
| Sequence Name     | fl         |
| Dimension         | 2D         |
| Excitation        | Slice-sel. |
| RF Pulse Type     | Normal     |
| Gradient Mode     | Normal     |
| Flow Compensation | None       |
| Bandwidth         | 320 Hz/Px  |
| Asymmetric Echo   | Allowed    |
| Segments          | 1          |

**Sequence - Part 2**

|                          |     |
|--------------------------|-----|
| Introduction             | On  |
| RF Spoiling              | On  |
| Acoustic noise reduction | Off |

**Sequence - Assistant**

|               |     |
|---------------|-----|
| SAR Assistant | Off |
| Allowed Delay | 0 s |

## \\Research\\Investigators\\Frederick\\extravals\\AAHead\_Scout\_64ch

TA: 14 sec Coil Selection: Manual Voxel Size: 1.6×1.6×1.6 mm<sup>3</sup> Acc:: 3 Rel. SNR: 1.00**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | Off                |
| Wait for User to Start                        | Off                |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | Off                |
| Auto Store Images                             | On                 |
| Disable auto transfer to PACS                 | Off                |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Routine**

|                     |                    |
|---------------------|--------------------|
| Slab Group          | 1                  |
| Slabs               | 1                  |
| Distance Factor     | 20 %               |
| Position            | L0.0 A10.0 H0.0 mm |
| Orientation         | Sagittal           |
| Phase Encoding Dir. | A >> P             |
| Slices per Slab     | 128                |
| Phase Oversampling  | 0 %                |
| Slice Oversampling  | 0.0 %              |
| FOV Read            | 260 mm             |
| FOV Phase           | 100.0 %            |
| Slice Thickness     | 1.6 mm             |
| TR                  | 3.15 ms            |
| TE                  | 1.37 ms            |
| Averages            | 1                  |
| Concatenations      | 1                  |
| AutoAlign           | Head               |
| Coil Elements       | HC1-7;NC1          |

**Contrast - Common**

|                    |           |
|--------------------|-----------|
| TR                 | 3.15 ms   |
| TE                 | 1.37 ms   |
| Flip Angle         | 8 deg     |
| Fat-Water Contrast | Standard  |
| Contrasts          | 1         |
| Reconstruction     | Magnitude |

**Contrast - Dynamic**

|                |          |
|----------------|----------|
| Dynamic Mode   | Standard |
| Measurements   | 1        |
| Time to Center | 6.2 s    |

**Resolution - Common**

|          |        |
|----------|--------|
| FOV Read | 260 mm |
|----------|--------|

**Resolution - Common**

|                  |           |
|------------------|-----------|
| FOV Phase        | 100.0 %   |
| Slice Thickness  | 1.6 mm    |
| Base Resolution  | 160       |
| Phase Resolution | 100 %     |
| Slice Resolution | 69 %      |
| Trajectory       | Cartesian |

**Resolution - Acceleration**

|                        |            |
|------------------------|------------|
| Acceleration Mode      | GRAPPA     |
| Reference Scans        | Integrated |
| Acceleration Factor PE | 3          |
| Reference Lines PE     | 24         |
| Acceleration Factor 3D | 1          |
| Phase Partial Fourier  | 6/8        |
| Slice Partial Fourier  | 6/8        |
| Asymmetric Echo        | Weak       |

**Resolution - Filter**

|                       |         |
|-----------------------|---------|
| Raw Filter            | Off     |
| Elliptical Filter     | Off     |
| Distortion Correction | 3D      |
| Normalize             | Prescan |
| Noise Masking         | Off     |
| Image Filter          | Off     |

**Geometry - Common**

|                     |                    |
|---------------------|--------------------|
| Slab Group          | 1                  |
| Slabs               | 1                  |
| Distance Factor     | 20 %               |
| Position            | L0.0 A10.0 H0.0 mm |
| Orientation         | Sagittal           |
| Phase Encoding Dir. | A >> P             |
| Slices per Slab     | 128                |
| Phase Oversampling  | 0 %                |
| Slice Oversampling  | 0.0 %              |
| FOV Read            | 260 mm             |
| FOV Phase           | 100.0 %            |
| Slice Thickness     | 1.6 mm             |
| TR                  | 3.15 ms            |
| Multi-Slice Mode    | Sequential         |
| Series              | Ascending          |
| Concatenations      | 1                  |

**Geometry - AutoAlign**

|                     |                    |
|---------------------|--------------------|
| Slab Group          | 1                  |
| Position            | L0.0 A10.0 H0.0 mm |
| Orientation         | Sagittal           |
| Phase Encoding Dir. | A >> P             |
| AutoAlign           | Head               |
| Initial Position    | Isocenter          |

**Geometry - AutoAlign**

|                     |             |
|---------------------|-------------|
| L                   | 0.0 mm      |
| P                   | 0.0 mm      |
| H                   | 0.0 mm      |
| Initial Orientation | Transversal |
| Initial Rotation    | 0.00 deg    |

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 0 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**System - Miscellaneous**

|                     |                  |
|---------------------|------------------|
| Coil Selection      | Manual           |
| Radial Sorting      | Off              |
| MSMA                | S - C - T        |
| Sagittal            | R >> L           |
| Coronal             | A >> P           |
| Transversal         | F >> H           |
| Coil Combination    | Adaptive Combine |
| Matrix Optimization | Off              |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Tune up  |
| B1 Shim               | TrueForm |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | Off      |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |             |
|-------------|-------------|
| Position    | Isocenter   |
| Orientation | Transversal |
| Rotation    | 0.00 deg    |
| A >> P      | 263 mm      |
| R >> L      | 350 mm      |
| F >> H      | 350 mm      |
| Reset       | Off         |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Non-sel. |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.249470 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Correction Factor   | 1.00           |
| Image Scaling       | 1.000          |

**Physio - PACE**

|                |     |
|----------------|-----|
| Resp. Control  | Off |
| Concatenations | 1   |

**Inline - Subtraction**

|                      |     |
|----------------------|-----|
| Subtract             | Off |
| Measurements         | 1   |
| StdDev               | Off |
| Save Original Images | On  |

**Inline - MIP**

|                      |     |
|----------------------|-----|
| MIP Sag              | Off |
| MIP Cor              | Off |
| MIP Tra              | Off |
| MIP Time             | Off |
| Radial MIP           | Off |
| Save Original Images | On  |
| MPR Sag              | Off |
| MPR Cor              | Off |
| MPR Tra              | Off |

**Inline - Composing**

|                  |     |
|------------------|-----|
| Inline Composing | Off |
|------------------|-----|

**Sequence - Part 1**

|                 |           |
|-----------------|-----------|
| Sequence Name   | fl        |
| Dimension       | 3D        |
| Excitation      | Non-sel.  |
| RF Pulse Type   | Fast      |
| Gradient Mode   | Normal    |
| Bandwidth       | 540 Hz/Px |
| Asymmetric Echo | Weak      |

**Sequence - Part 2**

|                    |           |
|--------------------|-----------|
| Introduction       | On        |
| RF Spoiling        | On        |
| Breast Application | Off       |
| Phase Enc. Order   | Automatic |

**Sequence - Assistant**

|               |     |
|---------------|-----|
| SAR Assistant | Off |
|---------------|-----|

## \\Research\\Investigators\\Frederick\\extravals\\X01

TA: 1:12 min Coil Selection: Auto Voxel Size: 2.5×2.5×2.5 mm<sup>3</sup> Acc:: 2 Rel. SNR: 1.00**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | Off                |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Disable auto transfer to PACS                 | Off                |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Routine**

|                          |              |
|--------------------------|--------------|
| Slice Group              | 1            |
| Slices                   | 60           |
| Distance Factor          | 0 %          |
| Position                 | Isocenter    |
| Orientation              | T > C-16.0   |
| Phase Encoding Dir.      | A >> P       |
| Phase Oversampling       | 0 %          |
| FOV Read                 | 216 mm       |
| FOV Phase                | 100.0 %      |
| Slice Thickness          | 2.5 mm       |
| TR                       | 4000.0 ms    |
| TE 1                     | 12.60 ms     |
| TE 2                     | 29.16 ms     |
| TE 3                     | 45.72 ms     |
| TE 4                     | 62.28 ms     |
| Averages                 | 1            |
| Multi-band accel. factor | 3            |
| AutoAlign                | Head > Brain |

**Contrast - Common**

|                    |                |
|--------------------|----------------|
| TR                 | 4000.0 ms      |
| TE 1               | 12.60 ms       |
| TE 2               | 29.16 ms       |
| TE 3               | 45.72 ms       |
| TE 4               | 62.28 ms       |
| MTC                | Off            |
| Magn. Preparation  | None           |
| Flip Angle         | 67 deg         |
| Fat-Water Contrast | Fat Saturation |
| Contrasts          | 4              |
| Reconstruction     | Magnitude      |

**Contrast - Dynamic**

|              |          |
|--------------|----------|
| Dynamic Mode | Standard |
| Measurements | 4        |

**Contrast - Dynamic**

|             |         |
|-------------|---------|
| Delay in TR | 0.00 ms |
|-------------|---------|

**Resolution - Common**

|                  |         |
|------------------|---------|
| FOV Read         | 216 mm  |
| FOV Phase        | 100.0 % |
| Slice Thickness  | 2.5 mm  |
| Base Resolution  | 86      |
| Phase Resolution | 72 %    |
| Interpolation    | Off     |

**Resolution - Acceleration**

|                        |             |
|------------------------|-------------|
| Acceleration Mode      | GRAPPA      |
| Reference scan mode    | Single-shot |
| Acceleration Factor PE | 2           |
| Reference Lines PE     | 24          |
| Phase Partial Fourier  | Off         |

**Resolution - Filter**

|                         |         |
|-------------------------|---------|
| Raw Filter              | On      |
| Elliptical Filter       | Off     |
| Hamming                 | Off     |
| Distortion Correction   | 2D      |
| Static Field Correction | Off     |
| Normalize               | Prescan |

**Geometry - Common**

|                          |             |
|--------------------------|-------------|
| Slice Group              | 1           |
| Slices                   | 60          |
| Distance Factor          | 0 %         |
| Position                 | Isocenter   |
| Orientation              | T > C-16.0  |
| Phase Encoding Dir.      | A >> P      |
| Phase Oversampling       | 0 %         |
| FOV Read                 | 216 mm      |
| FOV Phase                | 100.0 %     |
| Slice Thickness          | 2.5 mm      |
| TR                       | 4000.0 ms   |
| Multi-Slice Mode         | Interleaved |
| Series                   | Interleaved |
| Multi-band accel. factor | 3           |

**Geometry - AutoAlign**

|                     |                |
|---------------------|----------------|
| Slice Group         | 1              |
| Position            | Isocenter      |
| Orientation         | T > C-16.0     |
| Phase Encoding Dir. | A >> P         |
| AutoAlign           | Head > Brain   |
| Initial Position    | L0.0 P0.0 H6.0 |
| L                   | 0.0 mm         |
| P                   | 0.0 mm         |

**Geometry - AutoAlign**

|                     |          |
|---------------------|----------|
| H                   | 6.0 mm   |
| Initial Orientation | T > C    |
| T > C               | -16.00   |
| > S                 | 0.00     |
| Initial Rotation    | 0.00 deg |

**Geometry - Saturation**

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 6 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**System - Miscellaneous**

|                     |                  |
|---------------------|------------------|
| Coil Selection      | Auto Coil Select |
| Radial Sorting      | Off              |
| MSMA                | S - C - T        |
| Sagittal            | R >> L           |
| Coronal             | A >> P           |
| Transversal         | F >> H           |
| Coil Combination    | Sum of Squares   |
| Matrix Optimization | Performance      |
| Coil Focus          | Flat             |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Standard |
| B1 Shim               | TrueForm |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | On       |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |            |
|-------------|------------|
| Position    | Isocenter  |
| Orientation | T > C-16.0 |
| Rotation    | 0.00 deg   |
| A >> P      | 216 mm     |
| R >> L      | 216 mm     |
| F >> H      | 150 mm     |
| Reset       | Off        |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Standard |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.249470 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                          |           |
|--------------------------|-----------|
| 1st Signal/Mode          | None      |
| TR                       | 4000.0 ms |
| Log Signals              | Off       |
| Multi-band accel. factor | 3         |

**BOLD**

|                         |          |
|-------------------------|----------|
| GLM Statistics          | Off      |
| Ignore Meas. at Start   | 0        |
| Ignore After Transition | 0        |
| Model Transition States | On       |
| Temp. Highpass Filter   | On       |
| Threshold               | 4.00     |
| Paradigm Size           | 20       |
| Meas[1]                 | Baseline |
| Meas[2]                 | Baseline |
| Meas[3]                 | Baseline |
| Meas[4]                 | Baseline |
| Meas[5]                 | Baseline |
| Meas[6]                 | Baseline |
| Meas[7]                 | Baseline |
| Meas[8]                 | Baseline |
| Meas[9]                 | Baseline |
| Meas[10]                | Baseline |
| Meas[11]                | Active   |
| Meas[12]                | Active   |
| Meas[13]                | Active   |
| Meas[14]                | Active   |
| Meas[15]                | Active   |
| Meas[16]                | Active   |
| Meas[17]                | Active   |
| Meas[18]                | Active   |
| Meas[19]                | Active   |
| Meas[20]                | Active   |
| Motion Correction       | Off      |
| Spatial Filter          | Off      |
| Measurements            | 4        |
| Delay in TR             | 0.00 ms  |

**Sequence - Part 1**

|                   |             |
|-------------------|-------------|
| Sequence Name     | epfid       |
| Dimension         | 2D          |
| Excitation        | Standard    |
| Gradient Mode     | Performance |
| Flow Compensation | None        |
| Bandwidth         | 3230 Hz/Px  |
| Echo Spacing      | 0.50 ms     |
| Free Echo Spacing | Off         |
| EPI Factor        | 62          |

**Sequence - Part 2**

|              |     |
|--------------|-----|
| Introduction | Off |
| RF Spoiling  | Off |



**Sequence - Special**

|                           |          |
|---------------------------|----------|
| Excite pulse duration     | 3820 us  |
| Min. prep scans           | 0        |
| Min. prep scans SB        | 0        |
| Inter-TE delay            | 0 us     |
| Delay before PC scans     | 0 us     |
| Single-band images        | On       |
| MB LeakBlock kernel       | On       |
| MB dual kernel            | Off      |
| MB RF phase scramble      | On       |
| Opt. MB RF pulse BW       | Off      |
| SENSE1 coil combine       | On       |
| Invert RO/PE polarity     | On       |
| Echoes in separate series | Off      |
| Disable freq. update      | Off      |
| Suppress 16-bit DICOM     | Off      |
| Force equal slice timing  | Off      |
| Online multi-band recon.  | Online   |
| FFT scale factor          | 1.00     |
| Physio recording          | Off      |
| Triggering scheme         | Standard |

**Sequence - Assistant**

|               |     |
|---------------|-----|
| SAR Assistant | Off |
|---------------|-----|

## \\Research\\Investigators\\Frederick\\extravals\\X02

TA: 1:12 min Coil Selection: Auto Voxel Size: 2.5×2.5×2.5 mm<sup>3</sup> Acc:: 2 Rel. SNR: 1.00**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | Off                |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Disable auto transfer to PACS                 | Off                |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Routine**

|                          |              |
|--------------------------|--------------|
| Slice Group              | 1            |
| Slices                   | 63           |
| Distance Factor          | 0 %          |
| Position                 | Isocenter    |
| Orientation              | T > C-16.0   |
| Phase Encoding Dir.      | A >> P       |
| Phase Oversampling       | 0 %          |
| FOV Read                 | 216 mm       |
| FOV Phase                | 100.0 %      |
| Slice Thickness          | 2.5 mm       |
| TR                       | 4000.0 ms    |
| TE 1                     | 12.60 ms     |
| TE 2                     | 29.16 ms     |
| TE 3                     | 45.72 ms     |
| TE 4                     | 62.28 ms     |
| Averages                 | 1            |
| Multi-band accel. factor | 3            |
| AutoAlign                | Head > Brain |

**Contrast - Common**

|                    |                |
|--------------------|----------------|
| TR                 | 4000.0 ms      |
| TE 1               | 12.60 ms       |
| TE 2               | 29.16 ms       |
| TE 3               | 45.72 ms       |
| TE 4               | 62.28 ms       |
| MTC                | Off            |
| Magn. Preparation  | None           |
| Flip Angle         | 67 deg         |
| Fat-Water Contrast | Fat Saturation |
| Contrasts          | 4              |
| Reconstruction     | Magnitude      |

**Contrast - Dynamic**

|              |          |
|--------------|----------|
| Dynamic Mode | Standard |
| Measurements | 4        |

**Contrast - Dynamic**

|             |         |
|-------------|---------|
| Delay in TR | 0.00 ms |
|-------------|---------|

**Resolution - Common**

|                  |         |
|------------------|---------|
| FOV Read         | 216 mm  |
| FOV Phase        | 100.0 % |
| Slice Thickness  | 2.5 mm  |
| Base Resolution  | 86      |
| Phase Resolution | 72 %    |
| Interpolation    | Off     |

**Resolution - Acceleration**

|                        |             |
|------------------------|-------------|
| Acceleration Mode      | GRAPPA      |
| Reference scan mode    | Single-shot |
| Acceleration Factor PE | 2           |
| Reference Lines PE     | 24          |
| Phase Partial Fourier  | Off         |

**Resolution - Filter**

|                         |         |
|-------------------------|---------|
| Raw Filter              | On      |
| Elliptical Filter       | Off     |
| Hamming                 | Off     |
| Distortion Correction   | 2D      |
| Static Field Correction | Off     |
| Normalize               | Prescan |

**Geometry - Common**

|                          |             |
|--------------------------|-------------|
| Slice Group              | 1           |
| Slices                   | 63          |
| Distance Factor          | 0 %         |
| Position                 | Isocenter   |
| Orientation              | T > C-16.0  |
| Phase Encoding Dir.      | A >> P      |
| Phase Oversampling       | 0 %         |
| FOV Read                 | 216 mm      |
| FOV Phase                | 100.0 %     |
| Slice Thickness          | 2.5 mm      |
| TR                       | 4000.0 ms   |
| Multi-Slice Mode         | Interleaved |
| Series                   | Interleaved |
| Multi-band accel. factor | 3           |

**Geometry - AutoAlign**

|                     |                |
|---------------------|----------------|
| Slice Group         | 1              |
| Position            | Isocenter      |
| Orientation         | T > C-16.0     |
| Phase Encoding Dir. | A >> P         |
| AutoAlign           | Head > Brain   |
| Initial Position    | L0.0 P0.0 H6.0 |
| L                   | 0.0 mm         |
| P                   | 0.0 mm         |

**Geometry - AutoAlign**

|                     |          |
|---------------------|----------|
| H                   | 6.0 mm   |
| Initial Orientation | T > C    |
| T > C               | -16.00   |
| > S                 | 0.00     |
| Initial Rotation    | 0.00 deg |

**Geometry - Saturation**

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 6 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**System - Miscellaneous**

|                     |                  |
|---------------------|------------------|
| Coil Selection      | Auto Coil Select |
| Radial Sorting      | Off              |
| MSMA                | S - C - T        |
| Sagittal            | R >> L           |
| Coronal             | A >> P           |
| Transversal         | F >> H           |
| Coil Combination    | Sum of Squares   |
| Matrix Optimization | Performance      |
| Coil Focus          | Flat             |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Standard |
| B1 Shim               | TrueForm |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | On       |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |            |
|-------------|------------|
| Position    | Isocenter  |
| Orientation | T > C-16.0 |
| Rotation    | 0.00 deg   |
| A >> P      | 216 mm     |
| R >> L      | 216 mm     |
| F >> H      | 158 mm     |
| Reset       | Off        |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Standard |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.249470 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                          |           |
|--------------------------|-----------|
| 1st Signal/Mode          | None      |
| TR                       | 4000.0 ms |
| Log Signals              | Off       |
| Multi-band accel. factor | 3         |

**BOLD**

|                         |          |
|-------------------------|----------|
| GLM Statistics          | Off      |
| Ignore Meas. at Start   | 0        |
| Ignore After Transition | 0        |
| Model Transition States | On       |
| Temp. Highpass Filter   | On       |
| Threshold               | 4.00     |
| Paradigm Size           | 20       |
| Meas[1]                 | Baseline |
| Meas[2]                 | Baseline |
| Meas[3]                 | Baseline |
| Meas[4]                 | Baseline |
| Meas[5]                 | Baseline |
| Meas[6]                 | Baseline |
| Meas[7]                 | Baseline |
| Meas[8]                 | Baseline |
| Meas[9]                 | Baseline |
| Meas[10]                | Baseline |
| Meas[11]                | Active   |
| Meas[12]                | Active   |
| Meas[13]                | Active   |
| Meas[14]                | Active   |
| Meas[15]                | Active   |
| Meas[16]                | Active   |
| Meas[17]                | Active   |
| Meas[18]                | Active   |
| Meas[19]                | Active   |
| Meas[20]                | Active   |
| Motion Correction       | Off      |
| Spatial Filter          | Off      |
| Measurements            | 4        |
| Delay in TR             | 0.00 ms  |

**Sequence - Part 1**

|                   |             |
|-------------------|-------------|
| Sequence Name     | epfid       |
| Dimension         | 2D          |
| Excitation        | Standard    |
| Gradient Mode     | Performance |
| Flow Compensation | None        |
| Bandwidth         | 3230 Hz/Px  |
| Echo Spacing      | 0.50 ms     |
| Free Echo Spacing | Off         |
| EPI Factor        | 62          |

**Sequence - Part 2**

|              |     |
|--------------|-----|
| Introduction | Off |
| RF Spoiling  | Off |

**Sequence - Special**

|                           |          |
|---------------------------|----------|
| Excite pulse duration     | 3820 us  |
| Min. prep scans           | 0        |
| Min. prep scans SB        | 0        |
| Inter-TE delay            | 0 us     |
| Delay before PC scans     | 0 us     |
| Single-band images        | On       |
| MB LeakBlock kernel       | On       |
| MB dual kernel            | Off      |
| MB RF phase scramble      | On       |
| Opt. MB RF pulse BW       | Off      |
| SENSE1 coil combine       | On       |
| Invert RO/PE polarity     | On       |
| Echoes in separate series | Off      |
| Disable freq. update      | Off      |
| Suppress 16-bit DICOM     | Off      |
| Force equal slice timing  | Off      |
| Online multi-band recon.  | Online   |
| FFT scale factor          | 1.00     |
| Physio recording          | Off      |
| Triggering scheme         | Standard |

**Sequence - Assistant**

|               |     |
|---------------|-----|
| SAR Assistant | Off |
|---------------|-----|

## \\Research\\Investigators\\Frederick\\extravals\\X03

TA: 1:12 min Coil Selection: Auto Voxel Size: 2.5×2.5×2.5 mm<sup>3</sup> Acc:: 2 Rel. SNR: 1.00**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | Off                |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Disable auto transfer to PACS                 | Off                |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Routine**

|                          |              |
|--------------------------|--------------|
| Slice Group              | 1            |
| Slices                   | 60           |
| Distance Factor          | 20 %         |
| Position                 | Isocenter    |
| Orientation              | T > C-16.0   |
| Phase Encoding Dir.      | A >> P       |
| Phase Oversampling       | 0 %          |
| FOV Read                 | 216 mm       |
| FOV Phase                | 100.0 %      |
| Slice Thickness          | 2.5 mm       |
| TR                       | 4000.0 ms    |
| TE 1                     | 12.60 ms     |
| TE 2                     | 29.16 ms     |
| TE 3                     | 45.72 ms     |
| TE 4                     | 62.28 ms     |
| Averages                 | 1            |
| Multi-band accel. factor | 3            |
| AutoAlign                | Head > Brain |

**Contrast - Common**

|                    |                |
|--------------------|----------------|
| TR                 | 4000.0 ms      |
| TE 1               | 12.60 ms       |
| TE 2               | 29.16 ms       |
| TE 3               | 45.72 ms       |
| TE 4               | 62.28 ms       |
| MTC                | Off            |
| Magn. Preparation  | None           |
| Flip Angle         | 67 deg         |
| Fat-Water Contrast | Fat Saturation |
| Contrasts          | 4              |
| Reconstruction     | Magnitude      |

**Contrast - Dynamic**

|              |          |
|--------------|----------|
| Dynamic Mode | Standard |
| Measurements | 4        |

**Contrast - Dynamic**

|             |         |
|-------------|---------|
| Delay in TR | 0.00 ms |
|-------------|---------|

**Resolution - Common**

|                  |         |
|------------------|---------|
| FOV Read         | 216 mm  |
| FOV Phase        | 100.0 % |
| Slice Thickness  | 2.5 mm  |
| Base Resolution  | 86      |
| Phase Resolution | 72 %    |
| Interpolation    | Off     |

**Resolution - Acceleration**

|                        |             |
|------------------------|-------------|
| Acceleration Mode      | GRAPPA      |
| Reference scan mode    | Single-shot |
| Acceleration Factor PE | 2           |
| Reference Lines PE     | 24          |
| Phase Partial Fourier  | Off         |

**Resolution - Filter**

|                         |         |
|-------------------------|---------|
| Raw Filter              | On      |
| Elliptical Filter       | Off     |
| Hamming                 | Off     |
| Distortion Correction   | 2D      |
| Static Field Correction | Off     |
| Normalize               | Prescan |

**Geometry - Common**

|                          |             |
|--------------------------|-------------|
| Slice Group              | 1           |
| Slices                   | 60          |
| Distance Factor          | 20 %        |
| Position                 | Isocenter   |
| Orientation              | T > C-16.0  |
| Phase Encoding Dir.      | A >> P      |
| Phase Oversampling       | 0 %         |
| FOV Read                 | 216 mm      |
| FOV Phase                | 100.0 %     |
| Slice Thickness          | 2.5 mm      |
| TR                       | 4000.0 ms   |
| Multi-Slice Mode         | Interleaved |
| Series                   | Interleaved |
| Multi-band accel. factor | 3           |

**Geometry - AutoAlign**

|                     |                |
|---------------------|----------------|
| Slice Group         | 1              |
| Position            | Isocenter      |
| Orientation         | T > C-16.0     |
| Phase Encoding Dir. | A >> P         |
| AutoAlign           | Head > Brain   |
| Initial Position    | L0.0 P0.0 H6.0 |
| L                   | 0.0 mm         |
| P                   | 0.0 mm         |

**Geometry - AutoAlign**

|                     |          |
|---------------------|----------|
| H                   | 6.0 mm   |
| Initial Orientation | T > C    |
| T > C               | -16.00   |
| > S                 | 0.00     |
| Initial Rotation    | 0.00 deg |

**Geometry - Saturation**

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 6 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**System - Miscellaneous**

|                     |                  |
|---------------------|------------------|
| Coil Selection      | Auto Coil Select |
| Radial Sorting      | Off              |
| MSMA                | S - C - T        |
| Sagittal            | R >> L           |
| Coronal             | A >> P           |
| Transversal         | F >> H           |
| Coil Combination    | Sum of Squares   |
| Matrix Optimization | Performance      |
| Coil Focus          | Flat             |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Standard |
| B1 Shim               | TrueForm |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | On       |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |            |
|-------------|------------|
| Position    | Isocenter  |
| Orientation | T > C-16.0 |
| Rotation    | 0.00 deg   |
| A >> P      | 216 mm     |
| R >> L      | 216 mm     |
| F >> H      | 180 mm     |
| Reset       | Off        |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Standard |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.249470 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                          |           |
|--------------------------|-----------|
| 1st Signal/Mode          | None      |
| TR                       | 4000.0 ms |
| Log Signals              | Off       |
| Multi-band accel. factor | 3         |

**BOLD**

|                         |          |
|-------------------------|----------|
| GLM Statistics          | Off      |
| Ignore Meas. at Start   | 0        |
| Ignore After Transition | 0        |
| Model Transition States | On       |
| Temp. Highpass Filter   | On       |
| Threshold               | 4.00     |
| Paradigm Size           | 20       |
| Meas[1]                 | Baseline |
| Meas[2]                 | Baseline |
| Meas[3]                 | Baseline |
| Meas[4]                 | Baseline |
| Meas[5]                 | Baseline |
| Meas[6]                 | Baseline |
| Meas[7]                 | Baseline |
| Meas[8]                 | Baseline |
| Meas[9]                 | Baseline |
| Meas[10]                | Baseline |
| Meas[11]                | Active   |
| Meas[12]                | Active   |
| Meas[13]                | Active   |
| Meas[14]                | Active   |
| Meas[15]                | Active   |
| Meas[16]                | Active   |
| Meas[17]                | Active   |
| Meas[18]                | Active   |
| Meas[19]                | Active   |
| Meas[20]                | Active   |
| Motion Correction       | Off      |
| Spatial Filter          | Off      |
| Measurements            | 4        |
| Delay in TR             | 0.00 ms  |

**Sequence - Part 1**

|                   |             |
|-------------------|-------------|
| Sequence Name     | epfid       |
| Dimension         | 2D          |
| Excitation        | Standard    |
| Gradient Mode     | Performance |
| Flow Compensation | None        |
| Bandwidth         | 3230 Hz/Px  |
| Echo Spacing      | 0.50 ms     |
| Free Echo Spacing | Off         |
| EPI Factor        | 62          |

**Sequence - Part 2**

|              |     |
|--------------|-----|
| Introduction | Off |
| RF Spoiling  | Off |

**Sequence - Special**

|                           |          |
|---------------------------|----------|
| Excite pulse duration     | 3820 us  |
| Min. prep scans           | 0        |
| Min. prep scans SB        | 0        |
| Inter-TE delay            | 0 us     |
| Delay before PC scans     | 0 us     |
| Single-band images        | On       |
| MB LeakBlock kernel       | On       |
| MB dual kernel            | Off      |
| MB RF phase scramble      | On       |
| Opt. MB RF pulse BW       | Off      |
| SENSE1 coil combine       | On       |
| Invert RO/PE polarity     | On       |
| Echoes in separate series | Off      |
| Disable freq. update      | Off      |
| Suppress 16-bit DICOM     | Off      |
| Force equal slice timing  | Off      |
| Online multi-band recon.  | Online   |
| FFT scale factor          | 1.00     |
| Physio recording          | Off      |
| Triggering scheme         | Standard |

**Sequence - Assistant**

|               |     |
|---------------|-----|
| SAR Assistant | Off |
|---------------|-----|

## \\Research\\Investigators\\Frederick\\extravals\\X04

TA: 1:12 min Coil Selection: Auto Voxel Size: 2.5×2.5×2.5 mm<sup>3</sup> Acc:: 2 Rel. SNR: 1.00**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | Off                |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Disable auto transfer to PACS                 | Off                |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Routine**

|                          |              |
|--------------------------|--------------|
| Slice Group              | 1            |
| Slices                   | 60           |
| Distance Factor          | 50 %         |
| Position                 | Isocenter    |
| Orientation              | T > C-16.0   |
| Phase Encoding Dir.      | A >> P       |
| Phase Oversampling       | 0 %          |
| FOV Read                 | 216 mm       |
| FOV Phase                | 100.0 %      |
| Slice Thickness          | 2.5 mm       |
| TR                       | 4000.0 ms    |
| TE 1                     | 12.60 ms     |
| TE 2                     | 29.16 ms     |
| TE 3                     | 45.70 ms     |
| TE 4                     | 62.26 ms     |
| Averages                 | 1            |
| Multi-band accel. factor | 3            |
| AutoAlign                | Head > Brain |

**Contrast - Common**

|                    |                |
|--------------------|----------------|
| TR                 | 4000.0 ms      |
| TE 1               | 12.60 ms       |
| TE 2               | 29.16 ms       |
| TE 3               | 45.70 ms       |
| TE 4               | 62.26 ms       |
| MTC                | Off            |
| Magn. Preparation  | None           |
| Flip Angle         | 67 deg         |
| Fat-Water Contrast | Fat Saturation |
| Contrasts          | 4              |
| Reconstruction     | Magnitude      |

**Contrast - Dynamic**

|              |          |
|--------------|----------|
| Dynamic Mode | Standard |
| Measurements | 4        |

**Contrast - Dynamic**

|             |         |
|-------------|---------|
| Delay in TR | 0.00 ms |
|-------------|---------|

**Resolution - Common**

|                  |         |
|------------------|---------|
| FOV Read         | 216 mm  |
| FOV Phase        | 100.0 % |
| Slice Thickness  | 2.5 mm  |
| Base Resolution  | 86      |
| Phase Resolution | 72 %    |
| Interpolation    | Off     |

**Resolution - Acceleration**

|                        |             |
|------------------------|-------------|
| Acceleration Mode      | GRAPPA      |
| Reference scan mode    | Single-shot |
| Acceleration Factor PE | 2           |
| Reference Lines PE     | 24          |
| Phase Partial Fourier  | Off         |

**Resolution - Filter**

|                         |         |
|-------------------------|---------|
| Raw Filter              | On      |
| Elliptical Filter       | Off     |
| Hamming                 | Off     |
| Distortion Correction   | 2D      |
| Static Field Correction | Off     |
| Normalize               | Prescan |

**Geometry - Common**

|                          |             |
|--------------------------|-------------|
| Slice Group              | 1           |
| Slices                   | 60          |
| Distance Factor          | 50 %        |
| Position                 | Isocenter   |
| Orientation              | T > C-16.0  |
| Phase Encoding Dir.      | A >> P      |
| Phase Oversampling       | 0 %         |
| FOV Read                 | 216 mm      |
| FOV Phase                | 100.0 %     |
| Slice Thickness          | 2.5 mm      |
| TR                       | 4000.0 ms   |
| Multi-Slice Mode         | Interleaved |
| Series                   | Interleaved |
| Multi-band accel. factor | 3           |

**Geometry - AutoAlign**

|                     |                |
|---------------------|----------------|
| Slice Group         | 1              |
| Position            | Isocenter      |
| Orientation         | T > C-16.0     |
| Phase Encoding Dir. | A >> P         |
| AutoAlign           | Head > Brain   |
| Initial Position    | L0.0 P0.0 H6.0 |
| L                   | 0.0 mm         |
| P                   | 0.0 mm         |



**Geometry - AutoAlign**

|                     |          |
|---------------------|----------|
| H                   | 6.0 mm   |
| Initial Orientation | T > C    |
| T > C               | -16.00   |
| > S                 | 0.00     |
| Initial Rotation    | 0.00 deg |

**Geometry - Saturation**

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 6 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**System - Miscellaneous**

|                     |                  |
|---------------------|------------------|
| Coil Selection      | Auto Coil Select |
| Radial Sorting      | Off              |
| MSMA                | S - C - T        |
| Sagittal            | R >> L           |
| Coronal             | A >> P           |
| Transversal         | F >> H           |
| Coil Combination    | Sum of Squares   |
| Matrix Optimization | Performance      |
| Coil Focus          | Flat             |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Standard |
| B1 Shim               | TrueForm |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | On       |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |            |
|-------------|------------|
| Position    | Isocenter  |
| Orientation | T > C-16.0 |
| Rotation    | 0.00 deg   |
| A >> P      | 216 mm     |
| R >> L      | 216 mm     |
| F >> H      | 224 mm     |
| Reset       | Off        |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Standard |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.249470 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                          |           |
|--------------------------|-----------|
| 1st Signal/Mode          | None      |
| TR                       | 4000.0 ms |
| Log Signals              | Off       |
| Multi-band accel. factor | 3         |

**BOLD**

|                         |          |
|-------------------------|----------|
| GLM Statistics          | Off      |
| Ignore Meas. at Start   | 0        |
| Ignore After Transition | 0        |
| Model Transition States | On       |
| Temp. Highpass Filter   | On       |
| Threshold               | 4.00     |
| Paradigm Size           | 20       |
| Meas[1]                 | Baseline |
| Meas[2]                 | Baseline |
| Meas[3]                 | Baseline |
| Meas[4]                 | Baseline |
| Meas[5]                 | Baseline |
| Meas[6]                 | Baseline |
| Meas[7]                 | Baseline |
| Meas[8]                 | Baseline |
| Meas[9]                 | Baseline |
| Meas[10]                | Baseline |
| Meas[11]                | Active   |
| Meas[12]                | Active   |
| Meas[13]                | Active   |
| Meas[14]                | Active   |
| Meas[15]                | Active   |
| Meas[16]                | Active   |
| Meas[17]                | Active   |
| Meas[18]                | Active   |
| Meas[19]                | Active   |
| Meas[20]                | Active   |
| Motion Correction       | Off      |
| Spatial Filter          | Off      |
| Measurements            | 4        |
| Delay in TR             | 0.00 ms  |

**Sequence - Part 1**

|                   |             |
|-------------------|-------------|
| Sequence Name     | epfid       |
| Dimension         | 2D          |
| Excitation        | Standard    |
| Gradient Mode     | Performance |
| Flow Compensation | None        |
| Bandwidth         | 3230 Hz/Px  |
| Echo Spacing      | 0.50 ms     |
| Free Echo Spacing | Off         |
| EPI Factor        | 62          |

**Sequence - Part 2**

|              |     |
|--------------|-----|
| Introduction | Off |
| RF Spoiling  | Off |

**Sequence - Special**

|                           |          |
|---------------------------|----------|
| Excite pulse duration     | 3820 us  |
| Min. prep scans           | 0        |
| Min. prep scans SB        | 0        |
| Inter-TE delay            | 0 us     |
| Delay before PC scans     | 0 us     |
| Single-band images        | On       |
| MB LeakBlock kernel       | On       |
| MB dual kernel            | Off      |
| MB RF phase scramble      | On       |
| Opt. MB RF pulse BW       | Off      |
| SENSE1 coil combine       | On       |
| Invert RO/PE polarity     | On       |
| Echoes in separate series | Off      |
| Disable freq. update      | Off      |
| Suppress 16-bit DICOM     | Off      |
| Force equal slice timing  | Off      |
| Online multi-band recon.  | Online   |
| FFT scale factor          | 1.00     |
| Physio recording          | Off      |
| Triggering scheme         | Standard |

**Sequence - Assistant**

|               |     |
|---------------|-----|
| SAR Assistant | Off |
|---------------|-----|

## \\Research\\Investigators\\Frederick\\extravals\\X05

TA: 1:12 min Coil Selection: Auto Voxel Size: 2.5×2.5×3.0 mm<sup>3</sup> Acc:: 2 Rel. SNR: 1.00

## Properties

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | Off                |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Disable auto transfer to PACS                 | Off                |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

## Routine

|                          |              |
|--------------------------|--------------|
| Slice Group              | 1            |
| Slices                   | 60           |
| Distance Factor          | 0 %          |
| Position                 | Isocenter    |
| Orientation              | T > C-16.0   |
| Phase Encoding Dir.      | A >> P       |
| Phase Oversampling       | 0 %          |
| FOV Read                 | 216 mm       |
| FOV Phase                | 100.0 %      |
| Slice Thickness          | 3.0 mm       |
| TR                       | 4000.0 ms    |
| TE 1                     | 12.60 ms     |
| TE 2                     | 29.16 ms     |
| TE 3                     | 45.72 ms     |
| TE 4                     | 62.28 ms     |
| Averages                 | 1            |
| Multi-band accel. factor | 3            |
| AutoAlign                | Head > Brain |

## Contrast - Common

|                    |                |
|--------------------|----------------|
| TR                 | 4000.0 ms      |
| TE 1               | 12.60 ms       |
| TE 2               | 29.16 ms       |
| TE 3               | 45.72 ms       |
| TE 4               | 62.28 ms       |
| MTC                | Off            |
| Magn. Preparation  | None           |
| Flip Angle         | 67 deg         |
| Fat-Water Contrast | Fat Saturation |
| Contrasts          | 4              |
| Reconstruction     | Magnitude      |

## Contrast - Dynamic

|              |          |
|--------------|----------|
| Dynamic Mode | Standard |
| Measurements | 4        |

## Contrast - Dynamic

|             |         |
|-------------|---------|
| Delay in TR | 0.00 ms |
|-------------|---------|

## Resolution - Common

|                  |         |
|------------------|---------|
| FOV Read         | 216 mm  |
| FOV Phase        | 100.0 % |
| Slice Thickness  | 3.0 mm  |
| Base Resolution  | 86      |
| Phase Resolution | 72 %    |
| Interpolation    | Off     |

## Resolution - Acceleration

|                        |             |
|------------------------|-------------|
| Acceleration Mode      | GRAPPA      |
| Reference scan mode    | Single-shot |
| Acceleration Factor PE | 2           |
| Reference Lines PE     | 24          |
| Phase Partial Fourier  | Off         |

## Resolution - Filter

|                         |         |
|-------------------------|---------|
| Raw Filter              | On      |
| Elliptical Filter       | Off     |
| Hamming                 | Off     |
| Distortion Correction   | 2D      |
| Static Field Correction | Off     |
| Normalize               | Prescan |

## Geometry - Common

|                          |             |
|--------------------------|-------------|
| Slice Group              | 1           |
| Slices                   | 60          |
| Distance Factor          | 0 %         |
| Position                 | Isocenter   |
| Orientation              | T > C-16.0  |
| Phase Encoding Dir.      | A >> P      |
| Phase Oversampling       | 0 %         |
| FOV Read                 | 216 mm      |
| FOV Phase                | 100.0 %     |
| Slice Thickness          | 3.0 mm      |
| TR                       | 4000.0 ms   |
| Multi-Slice Mode         | Interleaved |
| Series                   | Interleaved |
| Multi-band accel. factor | 3           |

## Geometry - AutoAlign

|                     |                |
|---------------------|----------------|
| Slice Group         | 1              |
| Position            | Isocenter      |
| Orientation         | T > C-16.0     |
| Phase Encoding Dir. | A >> P         |
| AutoAlign           | Head > Brain   |
| Initial Position    | L0.0 P0.0 H6.0 |
| L                   | 0.0 mm         |
| P                   | 0.0 mm         |

**Geometry - AutoAlign**

|                     |          |
|---------------------|----------|
| H                   | 6.0 mm   |
| Initial Orientation | T > C    |
| T > C               | -16.00   |
| > S                 | 0.00     |
| Initial Rotation    | 0.00 deg |

**Geometry - Saturation**

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 6 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**System - Miscellaneous**

|                     |                  |
|---------------------|------------------|
| Coil Selection      | Auto Coil Select |
| Radial Sorting      | Off              |
| MSMA                | S - C - T        |
| Sagittal            | R >> L           |
| Coronal             | A >> P           |
| Transversal         | F >> H           |
| Coil Combination    | Sum of Squares   |
| Matrix Optimization | Performance      |
| Coil Focus          | Flat             |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Standard |
| B1 Shim               | TrueForm |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | On       |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |            |
|-------------|------------|
| Position    | Isocenter  |
| Orientation | T > C-16.0 |
| Rotation    | 0.00 deg   |
| A >> P      | 216 mm     |
| R >> L      | 216 mm     |
| F >> H      | 180 mm     |
| Reset       | Off        |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Standard |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.249470 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                          |           |
|--------------------------|-----------|
| 1st Signal/Mode          | None      |
| TR                       | 4000.0 ms |
| Log Signals              | Off       |
| Multi-band accel. factor | 3         |

**BOLD**

|                         |          |
|-------------------------|----------|
| GLM Statistics          | Off      |
| Ignore Meas. at Start   | 0        |
| Ignore After Transition | 0        |
| Model Transition States | On       |
| Temp. Highpass Filter   | On       |
| Threshold               | 4.00     |
| Paradigm Size           | 20       |
| Meas[1]                 | Baseline |
| Meas[2]                 | Baseline |
| Meas[3]                 | Baseline |
| Meas[4]                 | Baseline |
| Meas[5]                 | Baseline |
| Meas[6]                 | Baseline |
| Meas[7]                 | Baseline |
| Meas[8]                 | Baseline |
| Meas[9]                 | Baseline |
| Meas[10]                | Baseline |
| Meas[11]                | Active   |
| Meas[12]                | Active   |
| Meas[13]                | Active   |
| Meas[14]                | Active   |
| Meas[15]                | Active   |
| Meas[16]                | Active   |
| Meas[17]                | Active   |
| Meas[18]                | Active   |
| Meas[19]                | Active   |
| Meas[20]                | Active   |
| Motion Correction       | Off      |
| Spatial Filter          | Off      |
| Measurements            | 4        |
| Delay in TR             | 0.00 ms  |

**Sequence - Part 1**

|                   |             |
|-------------------|-------------|
| Sequence Name     | epfid       |
| Dimension         | 2D          |
| Excitation        | Standard    |
| Gradient Mode     | Performance |
| Flow Compensation | None        |
| Bandwidth         | 3230 Hz/Px  |
| Echo Spacing      | 0.50 ms     |
| Free Echo Spacing | Off         |
| EPI Factor        | 62          |

**Sequence - Part 2**

|              |     |
|--------------|-----|
| Introduction | Off |
| RF Spoiling  | Off |

**Sequence - Special**

|                           |          |
|---------------------------|----------|
| Excite pulse duration     | 3820 us  |
| Min. prep scans           | 0        |
| Min. prep scans SB        | 0        |
| Inter-TE delay            | 0 us     |
| Delay before PC scans     | 0 us     |
| Single-band images        | On       |
| MB LeakBlock kernel       | On       |
| MB dual kernel            | Off      |
| MB RF phase scramble      | On       |
| Opt. MB RF pulse BW       | Off      |
| SENSE1 coil combine       | On       |
| Invert RO/PE polarity     | On       |
| Echoes in separate series | Off      |
| Disable freq. update      | Off      |
| Suppress 16-bit DICOM     | Off      |
| Force equal slice timing  | Off      |
| Online multi-band recon.  | Online   |
| FFT scale factor          | 1.00     |
| Physio recording          | Off      |
| Triggering scheme         | Standard |

**Sequence - Assistant**

|               |     |
|---------------|-----|
| SAR Assistant | Off |
|---------------|-----|